Lecture 31.14: DBSCAN Clustering — Density-Based Spatial Clustering











📌 Topic Overview

Comprehensive conceptual derivation of DBSCAN (Density-Based Spatial Clustering of Applications with Noise). Covers why density-based clustering solves arbitrary non-convex geometries, hyperparameters Epsilon (eps) and MinPts, Core points, Border points, Noise outliers (-1), and density reachability.

📋 Key Learning Objectives

- Why K-Means fails on non-convex, non-spherical, arbitrary cluster geometries.
- DBSCAN Core Parameters: Epsilon (neighborhood radius) and MinPts (minimum density).
- Point Taxonomy: Core Points, Border Points, and Noise Outliers (-1).
- Concepts of Density-Reachability and Density-Connectivity.
- Key Advantage: Automatically discovering number of clusters and filtering noise.

🔍 In-Depth Conceptual Breakdown

- **DBSCAN Kyu Aaya?**: K-Means hamesha circular/spherical clusters dhoondhta hai aur har outlier point ko kisi na kisi cluster mein jabardasti ghusa deta hai. **DBSCAN** density (ghana-pan) par kaam karta hai — yeh kisi bhi arbitrary shape (rings, moons, snakes) ke clusters dhoondh leta hai aur outliers ko alag fek deta hai!

- **2 Core Parameters**:

- 1. ϵ (**Epsilon**): Kisi point ke charo taraf ka radius (circle ka daayra).

- 2. **MinPts** (`min_samples`): Ek dense cluster banane ke liye ϵ-neighborhood ke andar kam se kam kitne points hone chahiye.

- **Points Ki 3 Categories (Ma'am Ka Blackboard Diagram)**:

- 1. **Core Point (Mukhya Point)**: Agar kisi point ke ϵ-radius circle ke andar $\geq \text{MinPts}$ points maujood hain, toh wo **Core Point** kehlaata hai.
- 2. **Border Point (Kinare Ka Point)**: Agar kisi point ke circle mein MinPts se kam points hain, LEKIN wo kisi Core Point ke daayre mein aata hai, toh wo **Border Point** hota hai.
- 3. **Noise / Outlier Point**: Wo point jo na toh Core Point hai aur na hi kisi Core Point ke pados mein aata hai. Isko DBSCAN label `-1` deta hai!

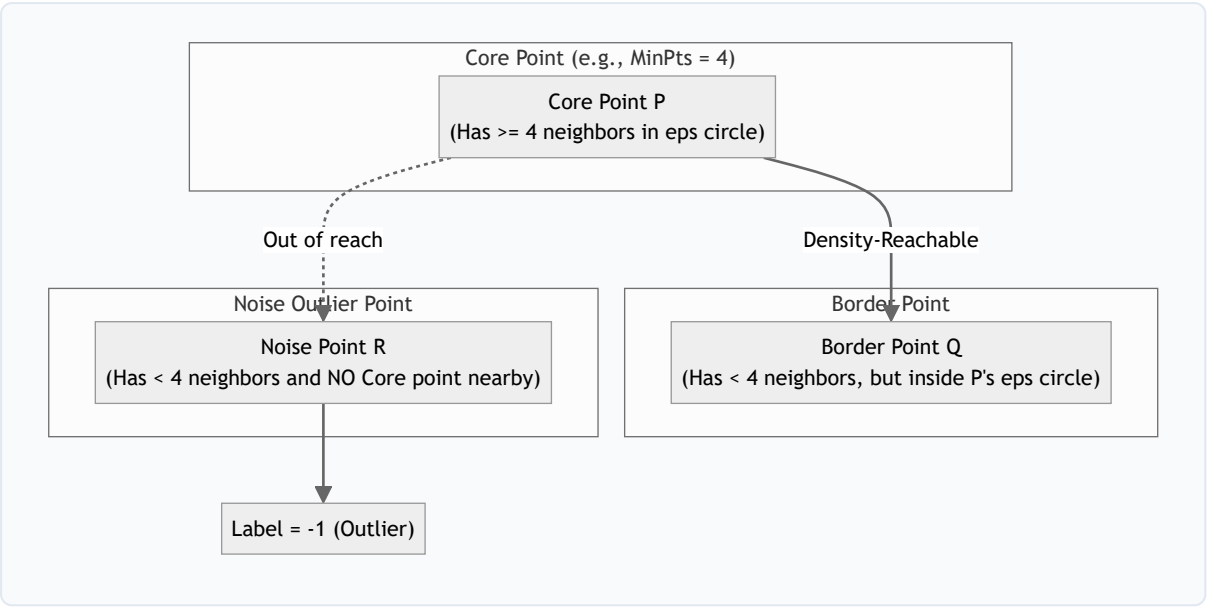
- **Cluster Formation (Density-Connectivity)**: Do Core Points agar ek doosre ke ϵ radius mein hain, toh unke clusters merge ho jaate hain. Is tarah point-to-point chain bante hue arbitrary shape ka lamba cluster ban jata hai.

- **Pure Theory Lecture**: Is lecture mein ma'am ne blackboard par Core, Border aur Noise points ke circles draw karke explain kiya hai, koi code nahi hai.

📋 DBSCAN Point Taxonomy Matrix

Point Category	Mathematical Condition	Neighborhood Density	Cluster Assignment
Core Point	$\lvert N_\epsilon(p) \rvert \geq \text{MinPts}$	Dense inner region	Seed of a cluster (expands cluster)
Border Point	$\lvert N_\epsilon(p) \rvert < \text{MinPts}$ and $p \in N_\epsilon(\text{Core})$	Marginal boundary density	Assigned to the neighboring Core cluster
Noise Outlier	$\lvert N_\epsilon(p) \rvert < \text{MinPts}$ and $p \notin N_\epsilon(\text{Core})$	Isolated sparse region	Assigned label `-1` (Filtered out)

📊 Ma'am's Blackboard Core, Border & Noise Circles



🔢 Mathematical Formulation

Epsilon-neighborhood definition for point p:

$$N_\epsilon(p) = \{q \in \mathcal{D} \mid d(p,q) \leq \epsilon\}$$

Condition for Core Point:

$$|N_\epsilon(p)| \geq \text{MinPts}$$

Density-Reachability condition: Point p is directly density-reachable from point q if:

$$p \in N_\epsilon(q) \quad \text{and} \quad |N_\epsilon(q)| \geq \text{MinPts}$$

📌 Note on Code

This is a pure conceptual, mathematical, and blackboard theory lecture. In accordance with course guidelines, no synthetic code is appended.